Solution Requirements Specification

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Functional Requirements (FR)

Req ID	Requirement Title	Detailed Specification & Acceptance Criteria
FR-01	Multimodal Evidence Ingestion	Accept JPG/PNG photos, WAV/MP3 audio, and MP4/AVI videos with client-side drag-and-drop validation.
FR-02	Real-Time Object & Weapon Detection	Execute RT-DETR object detection identifying humans, weapons, and evidence with confidence >= 0.70.
FR-03	3D Interactive Scene Reconstruction	Compute [X, Y, Z] spatial depth coordinates and render interactive Three.js 3D mesh with OrbitControls.
FR-04	Speech Transcription & Translation	Transcribe witness statements via Whisper and provide instantaneous English, Hindi, Telugu, and French tabs.
FR-05	OpenCV Video Evidence & CCTV Timeline	Extract frames at 1.0 Hz and run OpenCV computer vision detection for people, vehicles, objects, 2D bounding boxes, and 3D spatial trajectories.
FR-06	Resilient Forensic AI Assistant	Provide dual-tier conversational chatbot (Groq LLaMA-3.3 + Local Forensic Reasoning Engine) with one-click suggestion chips.
FR-07	Court-Admissible PDF Generation	Generate cryptographic SHA-256 validated PDF reports with evidence inventory, timeline, and legal certification.
FR-08	Acoustic Forensics & Waveform Player	Interactive audio player with play/pause, seek scrub slider, speed control, and animated 36-bar acoustic waveform visualizer.

2. Non-Functional Requirements (NFR)

Category	Specification	Implementation & Verification
Performance	< 3s for 2D image, < 5s for audio, < 8s for 15s video.	Local HuggingFace model cache + PyTorch inference optimization.
Durability	Zero data loss on restart or page reload.	Dual-tier session store persisting state as JSON records.
Security	Non-destructive read-only custody.	Werkzeug secure_filename and SHA-256 integrity validation.
Usability	Zero client install; runs in modern browsers.	Single Page Application built with HTML5, CSS3, and ES6 JS.